

UNIT - 3

APPROACHES OF TEACHING

3.1 Steps in planning a lesson

A key aspect of effective teaching is having a plan for what you are going to teach in the classroom. Creating such a Plan involves setting realistic goals deciding how to incorporate course, text books and resource materials and develop activities that will promote learning. The following points will help in planning a lesson.

Setting lesson goals

Lesson goals are most usually stated in terms of what students will achieve at the end of the lesson. Stating goals in this way allows both teacher and learners to know whether the goals have been achieved.

All goals must be SMART

- | | |
|----------------|--|
| S - Specific | Can the goal precise and well defined?
Is it clear?
Can every one understand it? |
| M - Measurable | What evidence is needed to confirm the completion?
How you will judge whether the goal have been attained or not? |
| A - Achievable | Is it within your capacity to achieve?
Are there sufficient resources are available to achieve? |
| R - Realistic | It is possible for you to perform the goal.
How does it fit in to your work? |
| T - Timed | Is there a deadline?
Is it feasible to attain in time? |

Example:

1. At the end of the lesson students will be able to find the area of a sector.
2. At the end of the lesson students will write all the subsets of a set.
3. At the end of the lesson students will be able to construct right angled triangle.
4. At the end of the lesson students will be able to draw a straight line.

3.2 Designing a Lesson plan

"Classrooms determine the India's future". Taking the importance of the above statement, lesson plan helps the teacher to plan and prepare classroom teaching. If a teacher goes to class without any preparation, he will not only waste the time but also face some problems. Hence to plan and prepare what to teach in each period, how to present it with systematic order and evaluation procedures to be done in that class room-all combining and presenting in short is called Lesson plan.

Lesson Plan Meaning

In lesson plan teacher has to clearly indicate, what are the teaching objectives and how he is going to teach the content in a specific teaching period. What are the aids to be used, what are the steps to use formative evaluation and what are the assessment are to be given etc. should be planned and noted in lesson plan.

Definitions of lesson plan

Good defines a lesson plan as an "outline of the important points of a lesson arranged in order in which they are to be presented to students by teacher."

Passing defines a lesson plan as an "organized statement of general and specific goals together with the specific means by which these goals are to be attained by the learner under the guidance of the teacher on the given day."

Standards, Lester B defines "a lesson plan is actually a plan of action. It includes the working philosophy of the teacher, her knowledge of philosophy, her information about the understanding of her pupils, her comprehension of the objectives of education, her knowledge of the material to be taught, and her ability to utilize effective methods."

Principles for preparing the Lesson plan

While preparing a lesson plan, the following principles should be kept in mind.

1. Students previous Knowledge, their capabilities must be basic for a lesson plan.
2. Present the subject matter in an organized, systematic and effective manner.
3. All objectives of lesson both general and specific should be known to the teacher.
4. Lesson plan should create interest in students.
5. Questions motivating self thinking, thought-provoking etc should find a place in lesson plan.

Need and Important features of a lesson plan

A good lesson have the following important characteristics.

1. All the objectives of the lesson both general and specific should be stated clearly in the lesson plan.
2. A good lesson plan should outline in detail the various steps that the teacher proposes to take in the class.
3. It prevents waste of time, as every step has been planned and written.

4. Lesson plan gives self confidence to the teacher.
5. A good lesson plan should indicate the list of AV aids to be used and should specify when and how to use them while teaching.
6. Sometimes we may give more time to unimportant subject matter. We can rectify this defect while preparing lesson plan.
7. A good lesson plan will integrate previous knowledge and experience to new knowledge.
8. A good lesson plan should stimulate independent thinking, interest, creative, imagination powers of the student.
9. A good lesson plan should be flexible and not rigid.
10. To complete the syllabus in time, lesson plan is needed.
11. It helps the teacher to ask proper and important questions to students.
12. It promotes a healthy learning environment.

Uses of Lesson Plan

1. Lesson plan helps the students to focus on the subject matter.
2. It helps to complete the lesson in time.
3. It prevents waste of time.
4. We can determine which Instructional Objectives have been fulfilled with the help of the lesson plan.
5. To become master in subject matter.
6. To get clarity in teaching activities.
7. To obtain teaching resources
8. While teaching it helps to avoid concept contradictions
9. It helps to prepare appropriate teaching aids
10. It brings self-confidence and self-power to teachers.

3.2.1 HERBARTIAN STEPS

German educator J.F. Herbart advocated five formal steps in teaching. They are

1. Preparation
2. Presentation
3. Association
4. Generalisation
5. Application

1. Introduction / Preparation

The introductory phase is called the 'warm-up' phase. The purpose is to create interest in students in the topic. The teacher may relate the new information to the one already taught or to the previous knowledge possesses by them. The new information is anchored with the existing knowledge.

2. Presentation

The transaction idea is done step by step following psychological approach. Starting from known things, the teacher leads the students to the new information by using comparisons, analogies, illustrations, demonstrations and other audio-visual aids including the chalk board.

3. Association

Through assimilation of the new idea through comparison with former ideas and consideration of their similarities and differences in order to in plant the idea in the mind.

4. Generalization

In science lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher

should guide the students only if their generalization is either incomplete or irrelevant.

5. Application

In this step the knowledge gained is applied to certain situations. This step is in conformity with the general desire of the students to make use of generalization in order to see for them where the generalizations are valid in some other situations.

3.2.2 Preparation of lesson plan

While preparing a lesson plan generally the following steps are followed:

1. Motivation / Preparation
2. Presentation
3. Association / Comparison
4. Generalisation
5. Application
6. Recapitulation
7. Assignment

1. Motivation / Preparation

Students are encourage and motivate to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

- Testing previous knowledge to learn new knowledge.
- Integrating the previous knowledge with the new knowledge.
- Arrest attention of the students using new techniques.
- Announcing the aim of the lesson.

2. Presentation:

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher

should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

3. Association / Comparison

Here the student understand the nature of the problem and compare the given information and analyse for the solution of the problem. Here knowledge is correlated with existing knowledge.

4. Generalization

In Mathematics lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application:

Here students apply the rules, formulas that they have learnt to solve the problems. By application of knowledge, the validity of generalization is tested and verified. Thus the knowledge becomes more meaningful to the learner and becomes permanent in his mind.

6. Recapitulation

Recapitulation helps the students to review the subject matter and consolidate ideas. Here Teacher will ascertain how well the students understood the new knowledge by way of asking few questions. From the answers we will also judge the effectiveness of the Teaching method.

7. Assignment / Home work

To learn mathematics effectively, class room activity alone is not enough. To know the mathematical concepts, laws, formulae and applications clearly we need home work

along with class room activities. Home work revise the portions done in the classroom. Assignments, motivate and reinforce further learning.

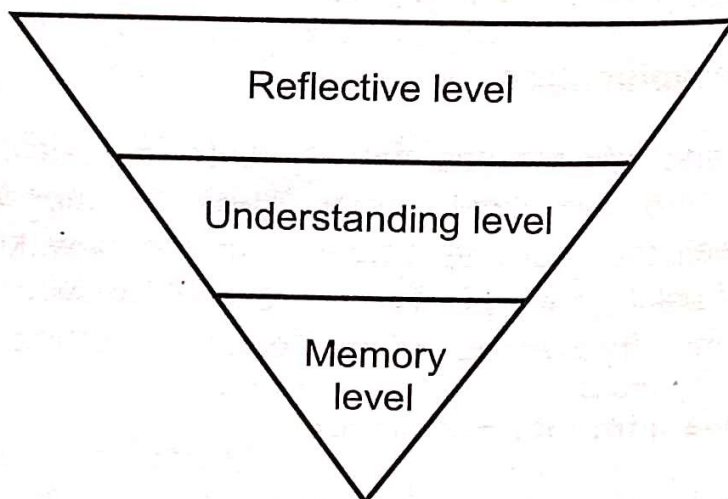
3.3 Organising Teaching

Organising of teaching-learning is the framework on which effective teaching is based. Careful and thoughtful planning allows instructional time to be maximised, standard to be addressed, prior knowledge to be activated, misconceptions to be controlled and the diverse characteristics and learning needs to be considered. Effective teachers use different methods for teaching to teach knowledge. In this topic we are going to see three different methods for teaching.

3.3.1 Memory level of Teaching (Herbatian Model)

Teaching is a part of teaching-learning process. It is required to bring specific changes in a person according to the need of his society. It should improve the learning skills of students. Teaching has to bring changes in the behaviour of students.

Levels of Teaching



Memory Levels of Teaching

Objectives

1. The objective of the memory level teaching is just to impart information or knowledge to the learners.
2. Retention and reproduction through recall or recognition.

In this level of teaching content, presentation and its exercise is important for learning. Emphasis is given on the presentation of information and facts in a systematic manner with a view to develop memory of students. At this level recall and recognition has important place.

Professor Herbart described the following five steps for the memory level of teaching. They are

- i. Focus
- ii. Syntax
- iii. Social system
- iv. Support system
- v. Evaluation system

We will discuss the above steps in detail.

i. Focus

Here importance is given on cramming of facts and development of following capacities.

- Training of mental aspects
- Providing knowledge of facts
- Retaining the learnt facts
- Recalling and Representing the learnt content

ii. Syntax

Herbert's five steps are

1. Preparation :

Important function of this step is to bring relevant ideas of the subject matter. The teacher plans about presentation of the contents.

2. Presentation :

The students are given new knowledge by testing their previous knowledge.

3. Comparison :

In this level teacher attempts to compare old facts and new facts, its similarities and dissimilarities, relationship between two subjects etc.

1. Generalization :

Pupils are given opportunities to think the content. Pupils formulate some principles and laws which can be used in the future life situations.

2. Application :

Teacher generate new situations to use learnt knowledge in new situation.

iii. Support System

Here supportive device are used to make learning more effective. The Audio visual aids are used to present the content in an observable form.

iv. Social System

This system consist of authoritarian behaviour. The teacher is more active to construct and present the subject matter systematically and logically in the classroom. The learners are passive listeners and they have to memorize and reproduce the content and their activities are directed by the teacher. At this level teacher provides motivation through

verbal praise or punishment. If the learner is motivated or reinforced continuously, then they learn better.

v. Evaluation system

Evaluation is the evidences about the realization of goals of teaching. Usually oral tests are given to measure the learnt material. The objective tests of recall and recognition types are given to evaluate the knowledge of the student.

Features of Memory Level

- Analysis of the subject matter is more important in this level of teaching. Much emphasis is given on the presentation of subject matter and not the experience of the students.
- Definite and systematic orders of subject matter is presented in this level of teaching.
- Teachers are not very flexible in the presentation of subject matter.
- The Mechanical handling of the various segments of content.
- Learning outcomes is of short duration.
- Interaction between teacher and students is at the lowest level.

Role of Teacher

- Dominating, active and authoritarian role
- He is the authority to instruct, deliver, direct, control and evaluate the performance.
- Takes initiatives, present the subject matter, directs the learner to do the activity in the desired manner.

- Right from the selection of the subject matter to its evaluation, the key process of teaching-learning remains with the teacher.

Suggestions for Memory Level Teaching

- Frequent recall or reproduction of content should be practised.
- Whole to part method should be used
- The content should have definite structure
- Teaching content should be integrated
- Contents should be presented systematically.
- Continuous reinforcement should be provided to the students.
- Students retention can be increased by practice.

Merits of Memory Level Teaching

1. Useful for children at lower classes.
2. The role of the teacher is important in this level of teaching and he is free to make choices of subject matter, plan it and can present it at will.
3. The knowledge acquired at memory level teaching forms a basis for the future.
4. Memory level teaching acts as the first step for understanding and reflective levels of teaching. It is pre-requisite for understanding level teaching.
5. It is useful for slow learners

Demerits of Memory Level Teaching

1. This does not contribute to the development of the students' capabilities.

2. At this level students learn by rote, so the knowledge gained does not prove helpful in real life situations as it does not develop the talents of students.
3. The pupils are kept in strict discipline.
4. Intelligence does not carry any importance in this type of teaching and it lacks motivation.
5. No interaction in the class
6. There is no guarantee of good retention and appropriate reproduction of the memorized material by the learner.

3.3.2 Understanding Level of Teaching (Morrison Model)

The teaching at the understanding level is of a higher quality than the one of the memory level. It is more useful and thoughtful from the point of view of mental capabilities. At this level of teaching, the teacher explains the student about the relationship between principles and facts and teach them how the principles can be applied. This method enables students to have complete command over subject material. No Cramming is encouraged here. The term understanding means to comprehend, to grasp, and to have knowledge of, to learn, to interpret and to infer etc.

Henry C Morrison clearly stated that understanding is not merely being able to recall something. It is not mere generalization deducted from specific facts; it is an insight into how it may be used in future situations. Morrison asserted that the outcome of all teaching is 'Mastery' and not memorization of facts. Morrison is the main promoter of understanding level promoter of understanding level of teaching.

Objectives :

1. Acquisition of required knowledge / facts
2. To see the relationship between acquired facts

Morrison has described five steps in his teaching model. They are

1. Exploration
2. Presentation
3. Assimilation
4. Organisation
5. Recitation

Let us discuss the above steps in detail

1. Exploration

In this step, the teacher tries to know the students 'entry-level behaviour' (Previous knowledge)

2. Presentation

The overview of the subject matter is given and the structure of the whole unit is discussed with the students.

3. Assimilation

In this step, the students study the subject matter deeply and try to related presented material to their internal self.

4. Organisation

The student present the acquired knowledge in a systematic manner without the help of teacher.

5. Recitation

This step is related to the students' verbal expression of the subject matter which is learned by them. They visit the whole content.

Morrison's Instructional Pattern

Morrison's teaching plan involves the following steps

1. Pretest

2. Teaching
3. Testing the result of instruction.
4. Changing the instruction procedures and
5. Teaching and testing again until the unit has been completely mastered by the student.

Subject matter and its presentation

The subject matter for the understanding level teaching is quite structured in terms of planning sequential and organised presentation and meaningful learning. It needs to be linked with the previous knowledge of the students and with its utilization for acquiring new facts and application to practical life.

Role of the teacher

Like memory level of teaching, the teacher plays a dominant and authoritarian role at the understanding level of teaching. This type of teaching is also subject centred and hence teacher has to pay his full attention in presenting the subject matter to his students to attain the desired understanding objectives besides the knowledge objective. He has to take care of the fact that the students gain complete mastery over the subject matter in term of its full understanding and generalized insight rather than its mere memorization. Lecture, Demonstration or explanation, question and answer method, inductive followed by deductive approach and analytic method with synthetic approach are all carried out at the understanding level.

Evaluation

Well-planned oral written and practical examinations are conducted in and outside the classroom. However emphasis is given to test to know ability to analyse, synthesize and

apply the generalised principles for deducing new facts and solving novel problems.

Merits of Understanding Level of Teaching

1. At this level of teaching students to make use of their thinking abilities.
2. Knowledge acquired at this level forms the basis of the reflective level of teaching
3. Here the teacher present subject matter before the students in an organized and sequential form. The new knowledge acquired is related to the previously acquired knowledge.
4. Here the students do not learn by rote. They learn by understanding the facts and information and their use and purpose.

Demerits of Understanding Level of Teaching

1. Teaching at this level is subject centred. There is no interaction between the teacher and students at this level.
2. In this type of teaching mastery is emphasized.
3. Ignore higher cognitive abilities.

3.3.3 Reflective Level of Teaching (Bigge and Hunt Model)

This level of teaching is the highest level of teaching - learning activity. Reflective level of teaching means, problem centred teaching. Hunt is the main promoter of reflective level of teaching. Reflective level teaching does not depend upon the memorization, understanding of concepts, and their application only; it demands the use of higher mental processes as

- Reasoning
- Thinking
- Analysing concepts
- Imagination
- ideas and thoughts

The learner will develop new insight to

- solve problems
- to discover
- to present generalization in the light of the new evidence
- to explore the field of knowledge
- to test and retest the achieved facts.
- This level basically involves the use of the scientific method.

Objectives

3. To develop insight into solve problems
4. To develop rational and critical thinking
5. To develop the ability of independent thinking and decision making.

Model of Reflective level of Teaching

Hunt developed the following steps for the Reflective level of Teaching. They are

a. Focus

To develop problem solving, critical and constructive, independent, original thinking

b. Syntax**Steps**

- i. Creating a problematic situation
- ii. Formulation of Hypothesis
- iii. Verify hypothesis
- iv. Collection of data
- v. Testing of hypothesis

c. Social system

Student occupies the primary place and teacher secondary place.

d. Support system

Objective type test is not used but essay type test is used. Attitude, beliefs and involvement is evaluated.

Features of Reflective level of teaching

- It is the highest level of teaching.
- It is problem - centric approach of teaching.
- The students are assumed to adopt some sort of research approach to solve the problem
- Classroom environment is independent. The learners are self-motivated and active.
- It leads to the development of creative skills.
- The students can solve problems of their lives by reasoning logic and imagination and lead successful and happy lives.
- Essay type test is used for evaluation. Attitude, belief and involvement are also evaluated.
- clear goals for pupil learning

Methods used for Reflective level of teaching

- Problem-solving method
- Investigating projects
- Heuristic method
- Experimental method
- Inquiry oriented method
- Analytic method

Merits of reflective level teaching

1. The teaching at this level is not teacher - centred or subject-centred. It is learner-centred.
2. There is an interaction between the teacher and the student at the reflective level teaching.
3. At this level, teaching is appropriate for the higher class.
4. At this level, teaching is highly thoughtful and useful then the teaching at the memory or understanding level.
5. This method is very useful for gifted children.

Demerits of reflective level teaching

1. It is not suitable for small children. It is suitable only for mentally matured children.
2. At this level, the study material is neither organised nor pre-planned. Therefore students cannot acquire systemic and organised knowledge of their courses.
3. It is a time consuming process.
4. It is not suitable for Indian class room.
5. It is not applicable for dull students.

3.4 Designing a Unit plan

Different meaning has been assigned to the term unit. They are i) The lesson of the day ii) As a chapter in a textbook etc. syllabus contains many topics / units such as Mensuration, Algebra, Triangles Trigonometry, Statistics etc.

Definition of a unit

Preston defines a unit as," a large block of related subject matter as can be over viewed by a learner."

Samford defines as," A unit is an outline of carefully selected subject matter".

Hence unit represents a careful organization of subject matter and learning experiences around a unifying principle.

For example the unit circle contains topics such as

1. Circle - definition and its elements
2. Area and circumference of a circle
3. Angle properties in a circle
4. Tangents to a circle

Characteristics of a Good unit

- A unit should be meaningful segments of well organized subject matter.
- Instructional objectives should be clearly defined in the unit plan.
- It should give new experience and Knowledge to students.
- It should help the individual differences in learning.
- A unit should not be too lengthily or too short.
- The length of the unit should be such as to retain the interest of the students.

Unit planning

A unit plan involves planning of teaching a unit, teaching methods, evaluation of teaching activities, diagnosing and remedial steps all together is called unit planning.

Proforma of a unit plan

Name of the Teacher :

Subject : Mathematics

Standard :

Unit :

Class and section :

Total no. of periods :

Sl. No.	Subject – Units	Content analysis	Objectives and specifications	Teaching and learning activities	Teaching aids used	Evaluation

Advantages of unit planning

1. Since we are dividing the unit into small parts and teach to the students, the students are able to understand the subject matter clearly.
2. Unit plan helps the teacher to teach the students in systematic manner.
3. Unit plan help the students to understand the relationship between the facts and principles that make up the unit.
4. Unit plan help us to study all information pertaining the unit.

5. Unit plan helps to teach according to individual difference of students.
6. It helps the teachers to prepare teaching aids.
7. It helps to develop self confidence in students.

Steps followed to perform unit plan.

We will see the steps to be followed for a better teaching - learning process in unit plan.

1. Analysis of content
2. State Instructional objectives
3. Instructional Resources Required
4. Planning the learning activities
5. Evaluation procedure
6. Remedial measures

1. Analysis of content

After selecting a unit, teacher has to analyse the contents of the unit in-depth. This helps the teacher to break up the unit into meaningful sub-units and retain the continuity throughout the unit.

2. Instructional Objectives

Knowledge of content skill learnt is analysed here.

3. Instructional Resources Required

Teaching aids required to deal the unit has to be planned and prepared well in advance.

4. Planning the learning activities

This part deals with the students activities in teaching learning process keeping in mind individual difference, the psychology of learning and objectives. The teacher also has to plan suitable teaching methods that are to be employed here.

5. Evaluation procedure

Teacher has to evaluate with reference to aims at the end of the unit or sub unit. Achievement test may be done and students progress and the effectiveness of teaching strategies may evaluate.

6. Remedial measures

After analyzing the achievement test of the students, take remedial measures if the performance is not satisfactory.

Method of Performing Unit Plan

Mathematics syllabus for the class may be divided into many units. Each unit is to be divided into small parts/lessons and then decide the number of periods to be allotted for each lesson. Plan all the units of the syllabus has to be completed in time. According to content of the unit divide it into required number of small parts require number and allot of periods.

Format to divide the syllabus into units

Class: 9

Unit No.	Name of the unit.	Topics.	No of periods required.	Remarks.
1.	Set Theory	Introduction, Explaining sets, methods to represent sets	3	Conducting Diagnosis Test
		Various kinds of sets, set operations, Ven diagram	7	
2.	Trigonometry	Introduction, Trigonometric ratios	3	Conducting Achievement Test

		Trigonometric ratios of some special angles, complementary angles	7	
		Using trigonometric tables	2	

Signature of the Teacher.

After dividing the syllabus into units and into small Topics, Prepare the Teaching plan for each Lesson/Topic and Complete it as per the time allotted to it.

Unit planning

A unit plan involves planning of teaching a unit, teaching methods, evaluation of teaching activities, diagnosing and remedial steps all together is called unit planning.

Content of a unit - Trigonometric ratios of some special angles

Sl. No.	Sub Division of the content	No. of periods required	Teaching Method	Resource Materials	Evaluation
1.	30°, 60° Trigonometric ratios	2	Deductive Method	1. Chart of Pythagoras theorem 2. Chart depicting Trigonometric ratios of 30°, 60°	
2.	Trigonometric ratios of 45°	1	Deductive Method	Chart depicting 45°	
3.	Trigonometric ratios of 0° and 90°	1	Deductive Method	Over head projector	

Sl. No.	Sub Division of the content	No. of periods required	Teaching Method	Resource Materials	Evaluation
4.	Solving problems with special angles	2	Analytic Method	Charts	
5.	Revision and test	1			At the end Achievement test will be conducted

3.4.1 Difference between lesson plan and unit plan.

Sr. No	Lesson Plan	Unit Plan
1.	Planned for a small portion to give effective teaching in one period.	Planned to teach a large content continuously for 6 to 10 periods or more.
2.	Lesson Plan is partial.	Unit plan is complete.
3.	Full concept in learning is not achieved.	Full concept in learning is achieved.
4.	Teaching alone will take place.	Teaching, evaluation, remedial teaching will take place.
5.	Questions will find a place in lesson plan to motivate self thinking, language ability etc.	There are no questions in unit plan.
6.	Assignments will find a place at the end.	There are no assignment.
7.	Lesson plan is written for each period.	According to length of the unit, unit plan is written for 6 to 10 periods or more.
8.	Method of teaching is not specified here.	Method of teaching is specified here.

3.5 Structure of a four-fold Lesson Plan

Introduction

We use the introduction to tell us a little about our lesson plan. Briefly describe the instructional techniques, what students to learn and any activities or assessments that you think are particularly noteworthy. The title of the lesson plan should be concise, clear and descriptive. It should invite teachers to take a closer look at the plan. Remember that teachers may see only the title and a short abstract of our plan in a page of search results.

Objectives

We should give some general and specific objectives of the teaching lesson. These objectives should be written clearly to achieve the learning outcomes. Learning outcomes are what students are expected to learn after completing the lesson plan.

Learning outcomes should be closely related to the curriculum alignment but should not simply repeat goals and objectives of the standard course of study. Learning outcomes may be broader, address particular aspects of curriculum objectives or teach the curriculum in a special context. Each learning outcome should be clearly reflected in the activities and assessed at the conclusion of the lesson.

Session time

Session time is the amount of time a teacher will need to schedule for the lesson plan. We might specify minutes, hours, class periods.

Materials

Materials needed include resources used by both teacher and student, including books, handouts, teaching aids, paper and pencils, art supplies and so on.

Previous knowledge of Learners

Write the previous knowledge required to understand the lesson plan.

Procedures

We should follow the following procedures to take the lesson.

1. Motivation
2. Preparation
3. Presentation
4. Review/Recapitulation
5. Follow up/Assignments

1. Motivation

- It could be through interesting introduction.
- The essence of introduction consists of arousing the interest of the pupil by supplying a motive for the study of a new unit.
- It is therefore better to call motivation.
- The essence of motivation consists in making the pupil understand what they are about and feel the worth wellness of studying it.

The most desirable motivation is for the pupil to see purpose in what he is doing. He must

- Recognize their need and its significance
- Identify the cause of his difficulties
- Have confidence in his ability to correct the difficulties
- Be given the proper exercise materials, which provide self-evaluation, and
- Be provided with opportunities for application of skills they have learnt

2. Preparation

- This is for laying the basic foundation for construction of the new lesson.
- Test the previous knowledge of the students, so that they could understand the present topic based on those previous knowledge/concepts.
- If the pre requisite concepts are not clear, they may not understand the present concept.

3. Presentation

- The most important part of teaching all skills learnt through micro teaching must be applied here to make the teaching effective.
- Make thorough content analysis-concept by concepts and then present the concept from simple to complex, from easy to difficult etc.
- Provide various activities group work, demonstration, explanation, questioning and blackboard work.

4. Review

- Provide the summary of concepts taught in that period.
- It may be fast restating by teacher or by asking questions and eliciting answers from students.
- This also will help to identify which part not understood well by students and brief explanation could be given.

5. Follow up

- It could be or simple assignment of asking to solve problems in exercise or
- Project work.

Students may be asked to observe from society or computer centers. It could be referring to nets/webs books etc.

The structure of a four-fold lesson plan is given below:

- Name of the Student Teacher : Date :
- Class / Section and session :
- Name of the school :
- Subject :
- Unit :
- Topic :
- Instructional Objectives : Write the list of objectives of the Topic to be covered. i.e. Write what the student knows at lesson completion.
- Instructional Resources Required : Write the list of Materials Aids required to explain the Topic
- Previous knowledge of Learners : Write the previous knowledge required to understand the new topic

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Preparing and giving motivation to receive new	Behavioural objectives are the smaller, observable understandable	Learning activities refers to all interaction, and practice	On the basis of content / concept and learning

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
knowledge / sequence of events that make up the lesson / Independent practice - Recapitulation	and measurable. It must describes learning outcomes	in which learning takes place in classroom. In this column teacher / student activities will take place to attain the objectives	experiences formative questions are framed here step by step.

Followup activities : Give homework and assignment in the topic covered.

Signature of Guide Teacher

Signature of the Student Teacher

3.6 Preparing of Model Lesson Plans

Model Lesson Plan - 1

Name of the Student Teacher : xxxx

Name of the Guide Teacher : xxxx

Class / Section and Session : X Std. A FN

Name of the school : xxxx

Subject : Mathematics

Unit : Mensuration

Topic : Total area of a cylinder

Date of teaching : xxxx

Instructional objectives**Student**

1. Acquires knowledge to find the total surface area of a cylinder.
2. tells how to find the total surface area of a cylinder.
3. Knowledge and understanding of total surface area of a cylinder is applied, to new and unfamiliar situations.
4. Develops skill to draw cylinder
5. Approaches cylinder with a scientific attitude
6. Solves the problem with an interest
7. Differentiate relation between surface area and total surface area of a cylinder.
8. Writes verbal statements into symbols.
9. Draws the figure of the cylinder neatly
10. Does calculations with speed and accuracy.

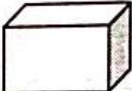
Instructional Resources Required

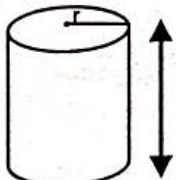
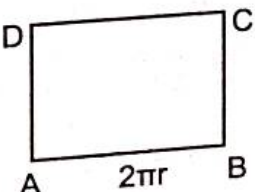
1. Cutouts of plane figures such as square, rectangle, triangle, circle etc.
2. Models of cube, sphere, cylinder
3. Rectangle sheet equivalent to surface area of the cylinder
4. Chart depicting cylinder

Previous Knowledge of learners :

Knowledge of Plane and Solid figures

Areas of Circle and Rectangle

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Recalls	The teacher shows some cut-outs of plane figures and test the previous knowledge  Square  Rectangle  Triangle  Circle	Tell the name of the cut-outs shown on the table
These figures have two dimensions	Compares	The teacher asks questions	How many dimensions to these figures have?
These figures are called plane figures.	Recalls		How are they called.
	Recalls	 Cube  Sphere  Cylinder	Tell the name of the solid objects shown on the table.
These objects have three dimensions	Compares	The teacher asks questions	How many dimensions do these objects have?
They are called as solid figures	Recalls		How are they called?
	Recalls	Teacher shows few objects and asked pupils to identify cylindrical objects Pencil, Tube light, Paint Tin	Which are the cylindrical shaped figures?
Introducing the topic of the lesson		Teacher announces the topic as "Total surface area of the cylinder" and writes the topic on the blackboard	

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
It is the two bases are circular it is called as circular cylinder. If the line joining centres of two equal circular base is perpendicular to the planes of the base, the cylinder is called as a right circular cylinder	Presentation	Teacher explains the meaning of right circular cylinder	What is the meaning of a right circular cylinder?
They are 3 surfaces for a cylinder. they are 1. The top circular base 2. The bottom circular base 3. The curved surface	Compares	The teacher shows a model of a cylinder and explains its surfaces	How many surfaces does a cylinder have?
Total surface area of a cylinder	Establishes relationship	The teacher shows a cylinder and asks few questions Total surface area of a cylinder = The curved surface area + area of top base + area of bottom base.	How you will find the total surface area of a cylinder?
	Acquires knowledge	Teacher explains with the help of chart drawn 	What is the radius and height of the cylinder?
Curved surface area of the cylinder is equal to the rectangular sheet of paper	Establishes relationship	The teacher fixes the rectangular sheet of paper on the surface area of the cylinder 	Curved surface area of cylinder is equal to whom?
	Compares	The teacher again wraps the rectangular sheet of paper on the cylinder and asks questions Length of rectangle	What is the length and breadth of the rectangle in terms of the dimensions of cylinder?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
		$AB =$ Circumference of the circular base of the Cylinder. Breadth of the rectangle $BC =$ height of the cylinder	
Circumference of a circle having radius r is $2\pi r$ units	Recalls	The teacher elicits the formula showing the bottom of the cylinder	What is the circumference of a circle having radius r ?
Surface area of a cylinder	Generalisation	The teacher elicits the steps involved to find the surface area of the cylinder and writes it on the blackboard Surface area of a cylinder = Area of the rectangle = Length $\times$ Breadth = $2\pi r \times h = 2\pi rh$ sq. units	What is the surface area of the cylinder in terms of radius and height of the cylinder?
Shape of top and bottom of cylinder is circle	Compares	The teacher elicits answer showing a cylinder	What is the shape of Top and Bottom of cylinder?
Area of circle = πr^2	Recalls	Teacher shows the circular bases of cylinder	What is the area of the circle?
Total surface area of the cylinder	Establishers relationships	Teacher explains with the help of the cylinder Total surface area of the cylinder = Surface area of the cylinder + area of the top + area of the bottom base $= 2\pi rh + \pi r^2 + \pi r^2$ $= 2\pi rh + 2\pi r^2$ $= 2\pi r(h+r)$ Sq. units	What is the total surface area of the cylinder?
Application : Circumference of a cylindrical base is 44 cm and its height is 12 cm. What is the total surface area of the cylinder?	Analysis	Teacher asks one of the student to read the problem.	What is given in the problem and what is asked?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Given circumference of the cylinder is 44 cm. Height of the cylinder is 12 cm. Asked : What is the total surface area of the cylinder?			
Radius of the base is needed	Checks the sufficiency of the data		What is more needed to find the total surface area of the cylinder?
	Establishes relationship?	Teacher explains how to find r. $2\pi r = 44 \text{ cm}$ $r = \frac{22}{2\pi}$ $= \frac{44 \times 7}{2 \times 22}$ $= 7 \text{ cm}$	How you will find the radius of the cylinder.
	Substitutes compute with speed and accuracy	Teacher asks one of the student to do the sum on the blackboard. Total surface area of a Cylinder = $2\pi r(h+r) = 2 \times \frac{22}{7} \times 7 \times (12+7)$ $= 44 \times 19 = 836$ Sq.units	What is the total surface area of the cylinder?
Teacher review the important points and asks few questions.	Recapitulation	1. What are the three surfaces of a cylinder? 2. What do you mean by right circular cylinder? 3. What is the formula to find the curved surface area of a cylinder? 4. What is the formula for total surface area of a cylinder?	

Follow up Activities

1. Find the total surface area of the cylinder whose diameter of the base is 14 cm and its height 25 cm.
2. Area of the base of a cylinder is 154 sq.cm. Its height is 10 cm. Find its curved surface area?

Signature of Guide Teacher

Signature of the Student Teacher

Model Lesson Plan - 2

Name of the Student Teacher : xxxx

Name of the Guide Teacher : xxxx

Class / Section and Session : IX Std. B AN

Name of the school : xxxx

Subject : Mathematics

Unit : Mensuration

Topic : Area of a sector (from l , r)

Date of teaching : xxxx

Instructional objectives**Student**

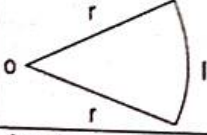
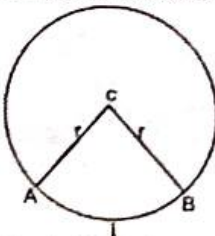
1. Acquires knowledge to find the area of a sector from l , r ?
2. tells how to find the area of a sector from l , r .
3. Knowledge and understanding applies area of a sector and applied to new and unfamiliar situations.
4. Develops skill to draw sector
5. Solves the problem with an interest.
6. Computes with speed and accuracy.
7. Solves the problem in an appropriate method.
8. Verifies the result.

Instructional Resources Required

1. Cutouts of circles, sectors.
2. Colour papers to cut in the shape of sectors.

Previous Knowledge of learners :

Knowledge of Circle, Sector, Angles of the Sector, Length of Arc of sector and Area of Sector.

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation:	Recognizes	The teacher shows the cut out of a circle It is circle	What is shape of this figure?
	Gives examples	Plate, bangles etc.	Give examples for circular things?
Part of a circle with two radius and an arc is called sector	Recalls	The teacher shows a cutout of a sector. 	What is a sector?
	Recalls	Angle of a sector and radius	What are required to find the area of a sector?
Angle of a Sector	Recalls	Teacher explains with a cutout of sector.  D = $\angle ACB$ is called angle of the sector.	What is meant by angle of a sector.
	Preparing models	Develops skill to cut a sector from the circle	Take a colour paper. Cut a circle with 6 cm radius and cut a sector from it.
	Recalls	Teacher elicits answer from the students. $A = \frac{D}{360} \pi r^2$	What is the formula to find area of the sector?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
	Recalls	Teacher elicits answer from the students. $l = \frac{D}{360} 2\pi r$	What is the formula to find the length if arc of a sector?
	Explains	$A = \frac{D}{360} 2\pi r \left(\frac{r}{2}\right)$ $= l \left(\frac{r}{2}\right)$	How can you modify the formula of area of sector if you want to introduce 'l' in it?
	Identifies	Teacher elicits answer from the students? $A = \frac{lr}{2}$	How can we tell the area of a sector now?
	Generalizes	Yes	Whether the formula we obtained to find the area of a sector is applicable to sectors.
Application: Area of a sector is 240π sq.cm. Its centre angle is 150° . Find the length of arc and radius of the sector.			
		The teacher asks one of the student to read the problem. Area of sector = 240π cm² $D = 150^\circ$ $A = \frac{D}{360} \pi r^2$ $240\pi = \frac{150}{360} \pi r^2$ $r^2 = \frac{240\pi \times 360}{150\pi}$ $= 576$ $r = \sqrt{576} = 24$ cm $l = \frac{D}{360} 2\pi r$ $= \frac{150}{360} 2\pi \times 24$ $= 20\pi$ cm	What is given in the problem?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Teacher review the important points and asks few questions.	Recapitulation	1. What is angle of a sector? 2. How you will find the use of a sector? 3. What is the formula to find the area of a sector? 4. What is the formula of the area if a sector in terms of l and r?	

Follow up Activities

1. Length of arc of a sector is 8π cm. Its radius is 10 cm. Find its area.
2. The centre angle of a sector is 140° and its radius is 9 cm. Find its length of arc.

Signature of Guide Teacher

Signature of the Student Teacher

Model Lesson Plan - 3

Name of the Student Teacher : xxxx

Name of the Guide Teacher : xxxx

Class / Section and Session : VIII Std. C AN

Name of the school : xxxx

Subject : Mathematics

Unit : Practical Geometry

Topic : Construction of parallelogram - Two adjacent sides and one diagonal is given.

Date of teaching : xxxx

Instructional objectives

Student

1. If two sides and one diagonal is given, student acquires the knowledge of the construction of parallelogram.
2. If two sides and one diagonal is given, student tells how to construct parallelogram.

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Teacher review the important points and asks few questions.	Recapitulation	1. What is angle of a sector? 2. How you will find the use of a sector? 3. What is the formula to find the area of a sector? 4. What is the formula of the area if a sector in terms of l and r?	

Follow up Activities

1. Length of arc of a sector is 8π cm. Its radius is 10 cm. Find its area.
2. The centre angle of a sector is 140° and its radius is 9 cm. Find its length of arc.

Signature of Guide Teacher

Signature of the Student Teacher

Model Lesson Plan - 3

Name of the Student Teacher : xxxx

Name of the Guide Teacher : xxxx

Class / Section and Session : VIII Std. C AN

Name of the school : xxxx

Subject : Mathematics

Unit : Practical Geometry

Topic : Construction of parallelogram - Two adjacent sides and one diagonal is given.

Date of teaching : xxxx

Instructional objectives

Student

1. If two sides and one diagonal is given, student acquires the knowledge of the construction of parallelogram.
2. If two sides and one diagonal is given, student tells how to construct parallelogram.

3. Knowledge and understanding applies to new and unfamiliar situations.
4. Gains skill is construction of parallelogram.
5. Draws the parallelogram with an interest.
6. Draws Parallelogram fairly.

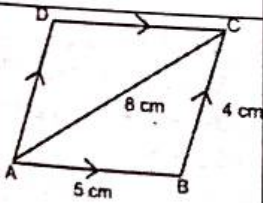
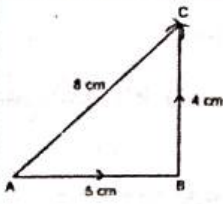
Instructional Resources Required

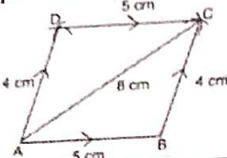
1. Cutouts of triangles, parallelogram and quadrilaterals and polygons.
2. Geometrical instrument box.

Previous Knowledge of learners :

Properties of a Parallelogram, Drawing a triangle, rectangle

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Identifies	The teacher shows a set of triangles, quadrilaterals and polygons.	Pick out the quadrilaterals from the top of the table.
Student identifies parallelogram	Identifies	Teacher asks a student to pick out parallelograms from quadrilaterals.	Pick out the parallelogram from the quadrilaterals.
Parallelogram	Recalls	1. Consist of two congruent triangles. 2. Two pairs of opposite sides are equal. 3. Two pairs of opposite interior angles are equal	What are the properties of a parallelogram.
Rectangle and squares are parallelograms	Compares	The teacher explains Rectangles, squares and rhombus and all parallelograms	Are rectangles and square are parallelograms?
To draw rectangle we need two measures. To draw square we need only one measurement.	Recall	The teacher shows the cut outs of rectangle and square	How many measurements need to draw rectangle, square?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
We need three measurements to draw a triangle	Recall	The teacher shows the cut out of a triangle.	How many measurements need to draw triangle.
Announcement of the topic	Gains knowledge	The teacher announces, we are going to see construction of parallelogram and writes it on the blackboard. We need three measurements	How many measurements need to draw a parallelogram?
The diagonal of a parallelogram divides the parallelogram into two congruent triangles.	Identifies	The teacher draws the parallelogram on the black board and ask a student to draw a diagonal.	How the diagonal of the parallelogram divides the parallelogram into?
	Recall	The teacher elicits the method of construction of a triangle from the students.	How you will draw a triangle if three measures are given?
	Judges the sufficiency	The teacher draws a rough figure of parallelogram and mark the measures 1. Draw $AB = 5\text{ cm}$ 2. With A and B as centres draw an arc of radii of 8 cm and 4 cm respectively. 3. Mark the point of intersection as C 4. Joint AC and BC 	Construct a parallelogram ABCD with $AB = 5\text{ cm}$, $AC = 8\text{ cm}$, $BC = 4\text{ cm}$

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
	Explains	With A and C as centres draw an arc of radii of 4 cm and 5 cm. Let the point of intersection as D. Join AD and CD ABCD is the required parallelogram 	How do you complete the parallelogram?
Area of the parallelogram = bh sq. units	Recall	b = base of the parallelogram h = distant between parallel lines	What is the formula to find the area of the parallelogram?
	Knowledge	Teacher explains with the help of set square and scale Draws perpendicular line from D to AB	How you will find the value of h ?
	Application	The teacher gives training to students to draw the parallelogram	Draw a parallelogram ABCD with $AB = 7$ cm $AC = 9$ cm and $BC = 5$ cm
Teacher explains the important points of parallelogram and put few questions on them	Recapitulation	1. How many measures needed to draw a parallelogram? 2. What are the properties of parallelogram? 3. How you will draw a parallelogram if two adjacent sides and diagonal is given 4. What is the formula to find the area of the parallelogram	

Follow up Activities

Draw a parallelogram PQRS with $PQ = 7$ cm $QR = 5$ cm and $PR = 9$ cm

Signature of Guide Teacher

Signature of the Student Teacher

Model Lesson Plan - 4

Name of the Student Teacher : xxxx

Name of the Guide Teacher : xxxx

Class / Section and Session : VII Std. B AN

Name of the school : xxxx

Subject : Mathematics

Unit : Real Number System

Topic : Addition of fractions

Date of teaching : xxxx

Instructional objectives

Student

1. Acquires knowledge to find the addition of fractions.
2. tells how to find the addition of fractions.
3. Knowledge and understanding of addition of fractions is applied, to new and unfamiliar situations.
4. Does calculations with speed and accuracy.
5. Verifies the result

Instructional Resources Required

1. Chart depicting fractions

Previous Knowledge of learners :

Knowledge of proper, improper and mixed fractions.

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Recalls	A proper fraction is of the form p/q where $p < q$ A Improper fraction is the form p/q where $p > q$	What is a proper fraction? What is a improper fraction?

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
		A Natural number and a proper fraction, combined into one number is called mixed fraction.	What is called a mixed fraction?
		Teacher interact with students and extract answers. Students are giving examples.	
Introducing the topic	Compares	Teacher announces the topic as "Addition of fractions" and write the topic on the Black board and explains	
Let us add two fractions $\frac{3}{7} + \frac{4}{7}$	Presentation	Teacher Explains How to add two natural numbers. Then in the same way explains for fractions shows the Chars of depictions fractions and explains.	What is $\frac{3}{7} + \frac{4}{7}$
	Recalls Acquires knowledge	LCM of 7, 7 is 7 $\frac{3}{7} + \frac{4}{7} = \frac{3+4}{7} = \frac{7}{7} = 1$	What is the LCM 7, 7
Letus add two fractions $\frac{3}{4} + \frac{1}{6}$	Acquired knowledge	LCM of 4, 6 is 12 $\frac{3}{4} + \frac{1}{6} = \frac{3 \times 3 + 1 \times 2}{12} = \frac{9+2}{12} = \frac{11}{12}$	What is the LCM of 4, 6
Simply $5\frac{1}{4} + 4\frac{3}{4} + 2\frac{5}{8}$	Recalls	$5\frac{1}{4} = \frac{5 \times 4 + 1}{4} = \frac{21}{4}$ $4\frac{3}{4} = \frac{4 \times 4 + 3}{4} = \frac{19}{4}$	What is the improper fraction of $5\frac{1}{4}$ What is the improper fraction of $4\frac{3}{4}$

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
		$2\frac{5}{8} = \frac{2 \times 8 + 5}{8} = \frac{21}{8}$	What is the improper fraction of $2\frac{5}{8}$
	Recalls	LCM of 4, 4, 8 is 8 $\begin{array}{r} 5\frac{1}{4} + 4\frac{3}{4} + 2\frac{5}{8} \\ \hline = \frac{21}{4} + \frac{19}{4} + \frac{21}{8} \end{array}$	What is the LCM of 4,4,8?
	Establish relationship	$= \frac{21 \times 2 + 19 \times 2 + 21 \times 1}{8}$ $= \frac{42 + 38 + 21}{8}$ $= \frac{101}{8} = 12\frac{5}{8}$	What is the mixed fraction of $\frac{101}{8}$?
Simplify $2\frac{3}{4} + 3\frac{5}{8} + 2\frac{1}{3}$	Application	Teacher ask a student to find the sum of the given problem in black board	
	Speed and accuracy	$\begin{array}{r} 2\frac{3}{4} + 3\frac{5}{8} + 2\frac{1}{3} \\ \hline = \frac{11}{4} + \frac{29}{8} + \frac{7}{3} \\ \hline = \frac{66 + 87 + 56}{24} \\ \hline = \frac{209}{24} = 8\frac{17}{24} \end{array}$	
Teacher review the important points	Recapitulation	Teacher review the important points and ask few questions.	1. What is the LCM of 5 and 4? 2. What is the mixed fraction of $\frac{165}{8}$.

Follow up Activities

Simplify

1. $\frac{3}{8} + \frac{4}{5} + \frac{1}{2}$

2. $3\frac{1}{4} + 2\frac{2}{3} + 4\frac{1}{12}$

Signature of Guide Teacher

Signature of the Student Teacher

Model Lesson Plan - 5

Name of the Student Teacher : xxxx

Name of the Guide Teacher : xxxx

Class / Section and Session : VI Std. A FN

Name of the school : xxxx

Subject : Mathematics

Unit : Geometry

Topic : Angles

Date of teaching : xxxx

Instructional objectives

Student

1. Acquires knowledge of angles
2. Defines what is an angle
3. Knowledge and understanding is applied to new and unfamiliar situation.
4. Identify the type of angles
5. Identify the angles formed by hour hand and minute hand in a clock.

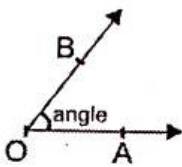
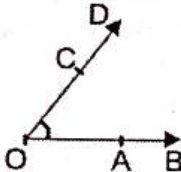
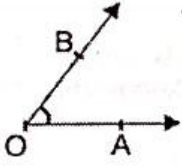
Instructional Resources Required

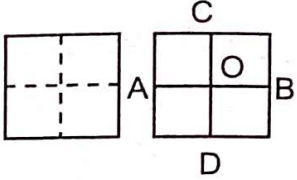
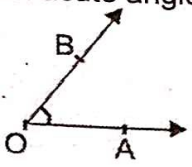
Charts depicting angle, right angle, Acute angle, Obtuse angle and Straight line angle.

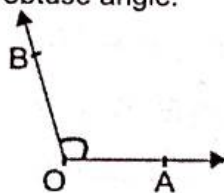
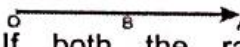
Previous Knowledge of learners :

Knowledge of point and Ray

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Motivation	Recalls	Make a point 'o' From O draw a ray $\overrightarrow{OA}$ 	What is the ray OA?

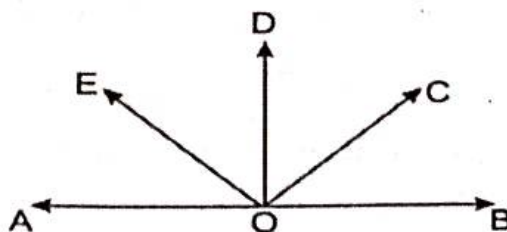
Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Introducing the Topic		Teacher announced the topic as, 'Angles' and write the topic on the blackboard.	
	Presentation	The figure formed by two rays with the same initial point is called an angle  The angle shown in the above figure is represented by $\angle AOB$ or $\angle BOA$ or sometimes as $\angle$	What is angle AOB
Angle	Identifies	Teacher shows a chart and ask questions $\angle AOC$ or $\angle BOC$ or $\angle BOD$ or $\angle AOC$	 Mention the angle in the above figure.
Measurer of the angle	Presentation Acquire Knowledge	 In the above figure let O as centre Ray $\overrightarrow{OA}$ rotates in the anticlockwise direction and reaches $\overrightarrow{OB}$. The rotation made by the ray is called the measure of that angle	

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Right angle	Presentation	 Fold a piece of paper as shown in the figure and unfold it. We get four angles. These angles are called right angles whose measure is 90°. Teacher explains with a chart. $\angle AOC$ or $\angle BOC$ or $\angle DOB$ or $\angle DOA$ are equal and their angle is 90°.	What is the measure of a right angle?
Straight line angle	Acquire knowledge	An angle whose measure is 180° is called a straight angle.  $\angle AOB = 180^\circ$ Teacher explains with a chart	What is the measure of a straight line angle?
Acute angle	Acquired knowledge	An angle whose measure is greater than 0 and less than 90° is called an acute angle.  $\angle AOB$ is an acute angle Teacher explains with a chart.	What is an acute angle.

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
	Understands	Angle 62° , Angle 85°	Give any two examples of acute angles
Obtuse angle	Acquires knowledge	An angle whose measure is greater than 90° and less than 180° is called an obtuse angle.  $\angle AOB$ is obtuse angle Teacher explains with a chart.	What is an obtuse angle?
	Understands	Angle 160° , Angle 95°	Give any two examples of obtuse angles.
Zero angle	Acquires knowledge	 If both the rays coincide the angle formed is 0° $\angle AOB = 0^\circ$	
Teacher review the important points and asks few questions	Recapitulation	1. What is the measure of right angle? 2. What is an acute angle? 3. What is an obtuse angle? 4. What is the measure of straight line angle?	

Follow up Activities

1. Give any two angles of obtuse angle.
2. Name the angles and write its kind.



Signature of Guide Teacher

Signature of the Student Teacher

Review Question

1. Define Lesson plan. Write the need and importance of Lesson Plan.
2. Explain the steps followed to prepare a lesson plan.
3. Explain in detail the Memory Level of Teaching. Write its Merits and Demerits.
4. Explain in detail the Understanding Level of Teaching. Write its Merits and Demerits.
5. Explain in detail the Reflective Level of Teaching. Write its Merits and Demerits.
6. Define a Unit Plan. What are the steps to be followed to perform a unit plan. Write its advantages.
7. Write the difference between Unit plan and Lesson plan.
8. Explain in detail the structure of a four-fold lesson plan.
9. Write a model lesson plan for IX Std. Mathematics.
10. Write a model lesson plan for VIII Std. Mathematics.